# Data 140 Lecture Notes

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# Chapter 1

## 1.1 Lecture 1

**Outcome space**: Set of all possible outcomes, denoted by $\Omega$. $\omega$ is a single outcome.

Experiment: Roll a die once. (six sided fair die). $\Omega = \{1, 2, 3, 4, 5, 6\}$

Event: Subset of outcome set (A, B, C, etc). Ex: Multiple of 3: $\{3, 6\}$

If the model is "equally likely outcomes", then P(multiple of 3) $= \frac{2}{6} = \frac{\#\text{ event}}{\#\Omega}$

### 1.1.1 Collisions in Hashing

$N$ codes, where $N \in \mathbb{Z}^+$, where there are $n$ individuals. A collision happens when 2 individuals are assigned the same code.

**Model** For each individual, pick one code uniformly at random from the $N$ codes, regardless of regardless of all other assignment (independtly).

$$P(\text{no collision}) = \frac{N!/n!}{N^n} = \prod_{i=0}^{n-2} \frac{N-i}{N}$$

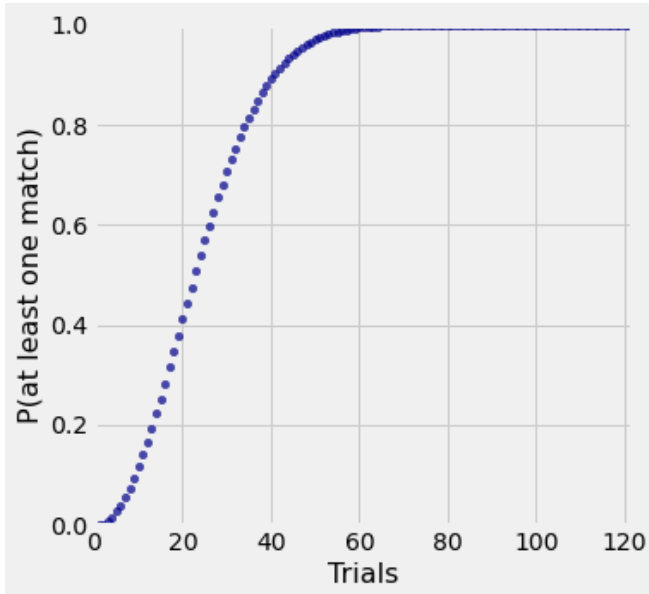
$$P(\text{at least one collision}) = 1 - P(\text{no collision}) = 1 - \prod_{i=0}^{n-1} \frac{N-i}{N}$$

A simple way to think about this is human birthdays.

Assumptions:

- Each person is equally likely to be born on each of the 365 days of the year. However, this is not actually true since we have lost leap years and bunching of birth dates, twins, triplets, etc

Under these simplifying assumptions. Compute.




$$
\begin{aligned}
\log(P(\text{no collision})) &= \sum_{i=0}^{n-1} \log(\frac{N-i}{N}) \\
&= \sum_{i=0}^{n-1} \log(1-\frac{i}{N}) \\
&\sim \sum_{i} \frac{-i}{N} \approx -\frac{1}{N}\sum i \approx -\frac{1}{N}\frac{(n-1)(n)}{2}
\end{aligned}
$$

For small i, $log(1+x) \sim x$

# Chapter 2

## 2.1 Lecture 02: Rules of Probability

1. $\Omega$ outcome space
2. Events are subsets of $\Omega$
3. Probability $P$ is a function on events $P(A)$

**Axioms of Probability:**

(Ground Rules)

1. $P(A) \geq 0 \forall A$ All probabilities are nonnegative
2. $P(\Omega) = 1$ The whole outcome space should be 1
3. Addition rule (union) $P(\cup_i A_i) = \sum_i P(A_i)$, given that $A_1, A_2, ...$ are mutually exclusive (don't overlap)

**Ex: Complements**

Taking the inverse of a subset. $P(A^c) + P(A) = P(\Omega)$. these build on the basic axioms

**Ex: Unions**

If we know the $P(\text{result} = 5) = P(\text{result} \leq 5) - P(\text{result} \leq 4)$

**Conditional Probability**

$$P(B|A) = \frac{P(AB)}{P(A)}$$

This works because if you say A occurs, then $\Omega = A$. Conditioning reduces your outcomes space. Then, since A has occurred then the $\omega = A \cap B$.

**Ex: Tickets**

Say you are drawing 2 tickets out of 3 without replacement. What is the probability that oyu get a blue and a red ticket?

$P(BR) = \frac{1}{6} = \frac{1}{2}$ of $\frac{1}{3} = P(\text{first B}) \cdot P(\text{second R}|\text{first B})$

### 2.1.1 Notation and Language

1. Outcome space: $\Omega$
2. Random Variable: $\Omega \rightarrow \mathbb{R}$
3.

**EX: A die rolled 5 times**

$\Omega = 6^5$, which is the results of 5 rolls of a die
Statistic $S$ : sum of the 5 numbers.
Function $\Omega \longrightarrow \{5, 6, 7, .., 30\}$

If we try to find

$$P(S > 15) = \sum_{k=16}^{30} P(S = k)$$

Which we read as, the chance that the sum exceeds 15 is the sum of the individual chances, where the results are the bounds [16,30].

Let $D_1$ = first roll, and $D_2$ = Second roll. Is $D_1 = D_2$?

$$\omega = [3, 1, 3, 4, 2]$$

Clearly, this shows that $D_1 \neq D_2$. They are different random variables. However, the distributions for $D_1$ and $D_2$ are the same. each value has the same uniform distribution. Therefore $D_1 \overset{d}{=} D_2$ in **distribution**

**Random Variables**

$X, Y$ are two random variables.

$$P(X = x, Y = y)$$

This means the probability that the random variable $X$ equals a given value $x$. The comma is an intersection, thus it is $X = x$ and $Y = y$.

If you sum over all possible $x$ and $y$, you get 1.

$$P(X = 1, Y = 4) = .0625$$

# Chapter 3

## 3.1 Lecture 04: 29 Jan 2026

Finished with basic concepts and methods for calculating probabilities. This lecture will be about applying these concepts in basic models. They are a set of assumptions about data we see in the world. Involve some elements of probabilities.

What makes a good model?

- reflects the setting of the experiment
- do not try to match every detail of that setting. No model that will pick up on everything. Pick up just enough that you can answer interesting questions

To illustrate a model, we have a Population of size $N$. This population is split into $G$ good and $B$ bad.

**Theorem 3.1.1** Hypergeometric Model

Population of size $N = G + B$
Good refers to something you care about, bad is everything else. These are not value statements or implications of morality. It is like Red or not Red.
From this population, get a simple random sample of size $n$. A SRS shuffle the population and take n. This is the same as sampling $n$ times without replacement. With this we can find something like $P(\text{g good elements in a sample})$

$$P(\text{g good elements in a sample}) = \frac{\binom{G}{g} \cdot \binom{B}{n-g}}{\binom{N}{n}}$$

The denominators is the total possible samples. The numerator is getting how many good elements, where we choose g good elements and have the rest bad. This is why we have the $n - g$. A quick check is that $G + B = N$ and $g + b = n$.
This equation is a hyper-geometric probability function. $(N, G, n)$. the numbers here are the fixed elements in the model. So in this model, we fix the number of elements, the number of good elements, and our sample size.

We would use this model when we draw without replacement, such as drawing from a deck of cards. Then, we also have to define what good means; does it mean that we get aces, we get a two pair, etc?

### 3.1.1 Sample with replacement:

**Example 3.1.1** (Roll a die)

Roll a die 5 time. What is the probability $P(2 \text{ sixes})$?
Intuitively, the chance of 2 sixes should be small. This is for a sanity check at the end.
We can think about this as 5 possible slots to fill with each trial. One way we can get a good event is

66321. we can also assume that probability of one roll is not affected by other roll. We can say that the probability that one happens is

$$\left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3$$

Then, we can count how many ways we can get this event; basically, out of 5 places how many times can we choose 2 places? Thus

$$P(2 \text{ sixes} = \binom{5}{2} \left(\frac{5}{6}\right)^2 \left(\frac{5}{6}\right)^3$$

**Theorem 3.1.2** Binomial Expansion

The new model for this is:

- $n$ independent Success Failure totals (S/F)
- each trial you have probability $p$ of $S$

$$P(\text{k successes}) = \binom{N}{k} p^k (1-p)^{n-k} \quad k = 0, 1, 2, \ldots, n$$

The key to this formula is the independence of the trials. Any pattern of k successes and n-k failures will have this formula.

This is the Binomial(n,p) distribution. It is used in the case when you have $n$ independent trials, and you are trying to find the probability of $k$ successes.

Checking this formula:

$$P(0 \text{ successes}) = \binom{n}{0} p^0 (1-p)^n - 0 = (1-p)^n$$

We want to examine this $\binom{n}{0}$

$$\binom{n}{0} = \frac{n!}{0!(n-0)!}$$

We define 0! = 1 is because of binomials! Define math on how you want to use them lol. The key here is the $\binom{n}{k}$ comes from he binomial expansion and pascal's triangle.

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} \tag{3.1}$$

We can take the binomial expansion, the binomial distribution function is $(p - (1-p))^n = 1^n = 1$.

**Note:-**

If running these calculations in Python, you can either do a sum of probability mass function (pdf), which evaluates how the probability $k$ successes, or cumulative distribution function (pdf), which evaluates the probability you get $\leq k$ successes

### 3.1.2 How to identify the mode of a discrete distribution

- $X$ binomial(n,p) distribution.
- P(k) = P(X=k)

This interprets to $X$ counting number of successes of $n$ independent success/failure trials, each of which has $p$ probability of success.

$$
\begin{aligned}
P(0) &= (1-p)^n \\
P(1) &= P(0) \cdot \frac{P(1)}{P(0)} \\
P(2) &= P(1) \cdot \frac{P(2)}{P(1)} = P(1) \cdot R(2).
\end{aligned}
$$

This is called the consecutive odds ratios.

$$R(k) = P(k) \cdot \frac{P(k)}{P(k-1)}, \quad k = 1, 2, \ldots, n$$

This is to build the probabilities from the R'S, instead of brute forcing calculations.

In the binomial distribution:

$$R(k) = \frac{\frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}}{\frac{n!}{(k-1)!(n-k+1)!} p^{k-1} (1-p)^{n-k+1}}$$

We do this because it is a lot easier to cancel things out.

$$R(k) = \frac{(n-k+1)p}{k(1-p)} = \left(\frac{n+1}{k} - 1\right) \frac{p}{1-p}$$

As $k$ increases, $R(k)$ decreases in $k$.

**Note:-**

Usually by convention we have $q = 1 - p$

If $R(k) = 1 \iff R(k) = R(k-1)$, $R(k) < 1 \iff R(k) < R(k-1)$, $R(k) > 1 \iff P(k) > P(k-1)$

this shows us that a decreasing function can only have 1 or 0 times to cross P = 1. So the binomial mode is the Largest $k$ so that $P(k) \geq P(k-1)$, because that is when it is rising.

This happens when

$$
\begin{aligned}
\left(\frac{n+1}{k} - 1\right) \frac{p}{q} &\geq 1 \\
\frac{n+1}{k} - 1 &\geq \frac{1-p}{p} \\
\frac{n+1}{k} &\geq \frac{1}{p} \\
k &\leq (n+1)p
\end{aligned}
$$

Thus, the mode is the integer part of $(n+1)p$

### 3.1.3 Situation where n large, $p_n$ small

When this happens, the natural thing is that $np_n \to$ number $\mu > 0$

In the binomial distribution

$$
\begin{aligned}
P(0) = (1-p_n)^n &\approx e^{-\mu} \\
\log(P(0)) = n \log(1 - P_n) &\sim -n(p_n) \approx -/mu
\end{aligned}
$$

$$P(1) \approx e^{-\mu} \cdot \frac{\mu}{1}$$

Basically, you can show that

$$R(k) \to \frac{\mu}{k} \text{as } n \to \infty$$

This is what is known as the Poisson distribution

# Chapter 4

## 4.1 Lecture 05: 3 February 2026

### 4.1.1 Extension of the binomial to more than two categories

**Example 4.1.1**

Let Population be: 10% of A, 30% of B, 60% of C.
Make 10 draws w/ replacement
$N_a$ number of A, $N_b$, $N_c$
What is $P(2A, 3B, 5C)$?

$$P(2A, 3B, 5C) = P(N_a = 2, N_b = 3, N_c = 5) = \binom{10}{2} \cdot \binom{8}{3} \cdot \binom{5}{5} \cdot 0.1^2 \cdot .3^3 \cdot .6^5$$

$$\frac{10!}{2!8!} \cdot \frac{8!}{3!5!} 0.1^2 \cdot .3^3 \cdot .6^5 = \frac{10!}{2!3!5!} .1^2 \cdot .3^2 \cdot .6^5$$

The first step is fixing the sequence e.g. $\{AABBBCCCCC\}$ Then, we need to find different ways to get similar sequences. So use 2 slots for As, 3 slots from the 8 remaining, 5 slots from the 5 remaining. The 8! cancels out.
This is called the **multinomial joint distribution**. It is the distribution of multiple variables. It is very close to the binomial distribution, with known number of draws, independence, and same population.

### 4.1.2 Poisson Distribution

One way this arises is a binomial large $n$ small $p \approx Poisson$ Parameter $\mu = np$. For example, in Binom(1000, $\frac{2}{1000}$). We would expect around 2 successes, which means $\approx Poisson(2)$. The mean of the Poisson distribution is a whole number, and the integer part of $\mu$ is a mode.

We should expect a lot of the probability in a Poisson to be near that parameter $\mu$.Which is used as a model for rare events.

When we compare Binom(1000, $\frac{2}{1000}$) and $Poisson(2)$, we can barely tell the difference. This is what it means to be an approximation.

Regardless, the Poisson is still a distribution in its own right, outside of the binomial distribution. It is not always binomial.

**Theorem 4.1.1** Poisson Distribution

$$X \text{ has } Poisson(\mu) \text{ distribution}$$

$$P(X = k) = e^{-\mu} \frac{\mu^k}{k!} c \quad k = 0, 1, 2, 3, \ldots$$

**Note:-**

This has infinitely many values. The $e^{-\mu}$ does not involve $k$, which is only there for one reason only. It is there to ensure that the function is a probability density function.
For example, if we do

$$\sum_{0}^{\infty} \frac{\mu^k}{k!} = e^{\mu}$$

from the Taylor series expansion, therefore we need the $e^{-\mu}$, we can show that the resultant product is 1.

We use the Poisson distribution because it models real life and has very nice properties. Therefore we can answer interesting questions simply.

**Example 4.1.2** (Sums)
$X \sim Poisson(\mu)$ $Y \sim Poisson(\lambda)$
With $X, Y$ independent.

$$S = X + Y$$

Find the Distribution of $S$

$$S = 0, 1, 2, 3, \ldots$$

$$P(S = s) = \sum_{k=0}^{s} P(X = k, Y = s - k)) = \sum_{k=0}^{s} e^{\mu} \frac{\mu^k}{k!} \cdot e^{-\lambda} \frac{\lambda^{s-k}}{(s-k)!}$$

**Note:-**

General Note: We can't have something on the RHS that is not on the DHS that remains after the calculations, therefore necessitating the sum.

We are now done with probability theory, and now onto algebra!

$$\begin{aligned}
&= e^{-(\mu+\lambda)} \frac{1}{s!} \sum_{k=0}^{s} \frac{s!}{k!(s-k)!} \mu^k \lambda^{s-k} \\
&= e^{-\mu+\lambda} \frac{1}{s!} \sum_{k=0}^{s} \binom{s}{k} \mu^k \lambda^{s-k} \\
&= e^{-(\mu+\lambda)} \frac{(\mu+\lambda)^s}{s!}
\end{aligned}$$

.

**Note:-**

We multiply by $\frac{s!}{s!}$ to make things easier

This gives us a new Poisson dist with a parameter $\mu + \lambda$. $S \sim Poission(\mu + \lambda)$. It makes intuitive sense because x is around $\mu$ and Y is around $\lambda$. Therefore S should be around $\mu + \lambda$

For a large $n$ in Poisson(n), e.g. 60, we will get a distribution that is roughly normal. This is because we add something that is very skewed, it becomes roughly normal when it is summed numerous times together.

**Theorem 4.1.2** Summation of a Poisson

$$Possion(X_0) + Poission(X_1) + \cdots + Poission(X_n) = Poission(nX)$$

Given that $X_1, X_2, \ldots, X_n$ are independent.

- $S$ number of successes
- $F$: number of failures

If the model is Binom$(n, p)$, $F = n - S$. $S, F$ is perfectly dependent.

If we randomize the number of trials

**Note:-**

Example process: Sampling from a population where there are a lot of categories. Draw a sample, and we want at least one from every category. What should the sample size should be?
If we randomize the sample size, the Poisson helps a lot.

If the model that the number of trials is a Poisson$(\mu)$ random variable $N$, then the behavior changes.

Given $N = n$ $S$ has the Binom$(n, p)$ distribution.

Given $N = 0$: $S = 0$ with probability 1.

Distribution of $S$

The conditional distribution of $S$ given $s$ is binomial. BUT NOT $S$.

$$s = 0, 1, 2, \ldots$$

There is no end to the number of trials because $N$ is a random variable. Therefore, there is no way that $S$ is binomial.

$$P(S = s) = \sum_{n=s}^{\infty} P(N = n, S = s)$$

Basically, this states that we need $n$ trials starting from $s$ successes, partitioning the event space

$$= \sum_{n=s}^{\infty} P(N = n) P(S = s \mid N = n)$$

$$= \sum_{n=s}^{\infty} e^{-\mu} \frac{\mu^n}{n!} \cdot \frac{n!}{s!(n-s)!} p^s q^{n-s}, \quad q = 1 - p$$

Probability theory is over....

We can cancel out the $n!$

$$e^{\mu} \frac{\mu^s p^s}{s!} \sum_{n=s}^{\infty} \frac{\mu^{n-s} q^{n-s}}{(n-s)!}$$

We split out the $\mu^s = \mu^{n-s} \cdot \mu^s$

$$e^{\mu} \frac{(\mu p)^s}{s!} \sum_{n=s}^{\infty} \frac{(\mu q)^{n-s}}{(n-s)!}$$

$$= e^{-\mu} \frac{(\mu p)^s}{s!} e^{\mu q}$$

$$e^{-\mu(1-s)} \frac{(\mu p)^s}{s!} = e^{\mu p} \frac{(\mu p)^s}{s!}$$

therefore have $Poisson(\mu p)$

$$P(S = s, F = f) = P(N = s + f, S = s) = e^{-\mu} \frac{\mu^{s+f}}{(s+f)!} \cdot \frac{(s+f)!}{(s!f!} p^s q^f$$

$$e^{-\mu p} \frac{(\mu p)^s}{s!} \cdot e^{-\mu q} \frac{(\mu q)^f}{f!}$$

Therefore S and F are independent through this property.

# Chapter 5

## 5.1 Lecture 06: 5 February 2026

### 5.1.1 Expectation

One of the most important parts of probability. When we develop these concepts, we should be thinking about why this concept has a name. Why is expected value defined? Because we refer to it constantly.

A distribution contains complete information about the behavior of our random variables. We can find the $P(X = x)$. Sometimes this is too much information. We only want to understand roughly how big something is occasionally, which often is finding the center of the distribution. In probability, this is the expectation or average.

**Definition 5.1.1**

Random Variable: $X$ has expected $E[X]$ $X : \Omega \to \mathbb{R}$ . Two equivalent definitions.
On the Domain space:

$$E[X] = \sum_{\forall \omega \in \Omega} X(\omega) \cdot P(\omega))$$

On the range space:

$$E[X] = \sum_{\forall x} x \cdot P(X = x)$$

Important: The average is not necessarily a valid value of random variable, but it is in the same units as the random variable. The expectation is a long run average value.

**Example 5.1.1** (toss 2 coins)

$$\omega = \{hh, ht, th, tt\}$$

let $x$ be the number of heads. model: fair coin.

$$x(\omega) = \{2, 1, 1, 0\}$$

$$p(\omega) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4}$$

$$x = \{2, 1, 0\}$$

on the domain space:

$$e[x] = 2 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = 1$$

On the range space:

$$E[X] = 2 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 0 \cdot \frac{1}{4} = 1$$

Typically the domain space definition helps us establish facts, and the range space helps us calculate.

### 5.1.2 Expected Value of Common Distributions

1. Constant: $P(X = c) = 1, E[X] = c$
2. Bern($p$). $E[X] = p$.
3. $X$ Uniform $\{1, 2, 3, \ldots, n\}$. $E[X] = \frac{n+1}{2}$
4. $X$ : Poisson($\mu$) $E[X] = \mu$
5. X: Geom(p) $E[X] = \frac{1}{p}$

**Note:-**

Bernoulli is very interesting, because we can define $I$ to be an indicator of $A$ .

$$I_A = \begin{cases} 0 \text{if A doesn't occur} \\ 1 \text{if A does occur} \end{cases}$$

Then $I_A \sim \text{Bern}(P(A))$
Thus $E[I_A] = P(A)$
We have thus shown that any probability is a distribution as well. The expectation of an indicator is the probability of that event.

**Example 5.1.2** (Finding expected value of Poisson)

Let $X \sim \text{Poisson}(\mu)$

$$\begin{aligned} E[X] &= \sum_{k=0}^{\infty} k \cdot P(X = k) \\ &= \sum_{k=0}^{\infty} k \cdot e^{-\mu} \frac{\mu^k}{k!} \\ &= 0 + \sum_{k=1}^{\infty} k \cdot e^{-\mu} \frac{\mu^k}{k!} \\ &= \sum_{k=1}^{\infty} e^{\mu} \frac{\mu^k}{(k-1)!} \\ &= e^{-\mu} \mu \sum_{k=1}^{\infty} \frac{\mu^{k-1}}{(k-1)!} \\ &= e^{-\mu} \cdot \mu \cdot \sum_{k=2}^{\infty} \frac{\mu^k}{k!} \\ &= e^{-\mu} e^{\mu} \cdot \mu \\ &= \mu \end{aligned}$$

**Definition 5.1.2: Geometric Distribution**

Often, this is number of times until event $X$ happens.
i.i.d. Bern($p$) sequence means independently and identically distributed sequence of 0 and 1s. AKA a sequence of coin tosses that lands heads with probability $p$.
Let $X$ be the number of trials til the first success. In this class "til" means up to and **until** the success (inclusive).

$$P(X = k) = q^{k-1}p \quad k = 1, 2, 3, \ldots$$

This is called

$$X \sim geometric(p)$$

$$P(X > k) = q^k$$

We can understand this by knowing the definition of Geometric distribution, because any success after $k$ means that everything else were failures.
We can use the tail sum formula to derive the expected value here.

$$E[X] = 1 + q + q^2 + q^3 + \cdots = \frac{1}{1-(1-p)} = \frac{1}{p}$$

Alternatively, we can use conditioning! (173 knowledge)

$$E[X] = 1 + p \cdot P(X - 1|x = 0) + q \cdot P(X - 1|x > 1) = 1 + 0 + qE[X] = \frac{1}{p}$$

**Definition 5.1.3: Tail sum formula for E(Y) with non neg integer valued Y**

Let $Y$ with values $0, 1, 2, \ldots$ $p_i = P(Y = i)$

$$\begin{aligned} E[Y] = p_1 & \\ + p_2 + p_2 & \\ + p_3 + p_3 + p_3 & \qquad = P(Y > 0) + P(Y > 1) + P(Y > 2) + \ldots \\ . & \end{aligned}$$

$$E[Y] = \sum_{k=0}^{\infty} P(Y > k)$$

### 5.1.3 Proprieties of Expectation

Function of X

Case 1.) Linear

Let $Y = aX + b$ Then $E[Y] = aE[X] + b$.

**Example 5.1.3**

$H$ is the number of heads in 20 tosses (biased coin). $T$ number of tails. Know $E[H] = 6$ . What is $E[T]$?.

$$T = 20 - H, E[T] = 20 - E[H]$$

Case 2.) Nonlinear.

Expectation of $Y = g(X)$. $E[Y] = \sum_{\forall x \in X} g(x) \cdot P(X = x)$

**Example 5.1.4**

$X = \{1, 0, 1\}$ with chances $P(X = k) = \frac{1}{3}$

$$E[X] = 0$$

$Y = X^2$

The range of $Y$ changes with the squared term. Therefore it has $P(Y = 1) = \frac{2}{3}$ and $P(X = 0) = \frac{1}{3}$.

$$E[Y] = 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}$$

This is the domain space definition.

**Example 5.1.5**

$$X \sim \text{Poisson}(\mu), E[X] = \mu$$

Find $E[X(X-1)]$

$$\begin{aligned} E[X(X-1)] &= \sum_{k=0}^{\infty} k(k-1) \cdot e^{-mu} \frac{\mu^k}{k!} \\ &= \sum_{k=2}^{\infty} e^{-\mu} \frac{\mu^k}{(k-2)!} \\ &= e^{-\mu} \cdot \mu^2 \sum_{k=2}^{\infty} \frac{\mu^{k-2}}{(k-2)!} \\ &= e^{-\mu} \mu^2 e^{\mu} \\ &= \mu^2 \end{aligned}$$

.

# Chapter 6

## 6.1 Lecture 07: 10 February 2026

### 6.1.1 Properties of Expectation

The expectation is a feature of the probability distribution. It is the point of the histogram where it balances. Two R.V equals in distributions, will be equal in expectation.

The expectation isn't always a mode or a place where there are a lot of probability. Balances at the middle point, which means that we need to look at variability as well.

**Definition 6.1.1: Linearity of Expectation**

$$X : \Omega \to \mathbb{R}$$
$$Y : \Omega \to \mathbb{R}$$
$$S = X + Y$$

If you need to talk about the sum of two functions, both functions have to have the same domain.

$$S(\omega) = X(\omega) + Y(\omega)$$

Apply the domain space definition of expectation.

$$\sum_{\text{all } \omega} S(\omega)P(\omega) = \sum_{\forall \omega} X(\omega)P(\omega) + \sum_{\forall \omega} Y(\omega)P(\omega)$$

By definition, this is:

$$E(S) = E(X) + E(Y)$$

**Example 6.1.1** (Expectation of the Binomial)

$$X \sim \text{Binom}(n, p)$$

X is the number of successes in n i.i.d. Trials with probability p of success on each single trial.
We are going to say $X = I_1 + I_2 + I_3 + \dots$ Where $I_j$ is 1 if one trial was a success, and 0 otherwise.

**Note:-**

MAKE SURE TO DEFINE THE INDICATOR

$$E(X) = \sum_{\forall j} E[I_j] = p + p + p + \cdots + p = np$$

We do not use independence anywhere in here.

**Example 6.1.2** (Expectation of hypergeometric)

$X : hypergeometric(N, G, n)$

X is the number of good elements in a SRS of size $n$ elements, drawn of a population of $N$ elements, of which $G$ is good.

$$X = I_1 + I_2 + I_3 + \cdots + I_n$$

Where $J_j$ is the indicator of good at Trial j

$$E[X] = E[I_1] + E[I_2] + \cdots + E[I_n]$$

This DOES NOT require independence, which is good because indicator variables are not i.i.d.

$$E[X] = \frac{G}{N} + \frac{G}{N} + \frac{G}{N} + \cdots = n\frac{G}{N}$$

This is the same thing as the binomial, because $\frac{G}{N}$ is the probability of success on a single trial. Its just that the nature of the dependence is different.

We only really see the difference when looking at the variance. Hyper geometric is less variable, because with/without replacements.

The method of indicators in these two examples are very important. We can split up the random variables into indicators, then we find the expectation of one indicator. The number of indicators to use is the amount of possible successes we can have. We have to make sure to define the indicator to make sure our expectation is correct.

### 6.1.2 Estimating a parameter

In data science, we try to estimate unknowns.

Notation: numerical parameter $\theta$. Estimate is a statistic $T$, where statistic is a function of the sample.

It is the number we calculate using the elements of the sample. We think $\theta$ is around this number. Then we have to ask if $T$ is any good.

Sample: $X_1, X_2, \ldots, X_n$ identically distributed.

Each element in this same have the same mean $\mu$.

$S_n = \sum X_i$, $E[S_n] = n\mu$.

This is a special case of the binomial expectation. They are a sum of Bernoulli R.Vs. the Key is that

$$\bar{X}_n = \frac{1}{n}S_n, E[\bar{X}_n = \frac{1}{n}E[S_n] = \mu]$$

The expectation of the sample mean is the population mean. On average, the sample gets the population right. This motivates the definition of unbiased estimates.

**Definition 6.1.2: Biased and Unbiased Estimates**

Bias of $T$: $E[T] - \theta$.

On average, how different is $T$ from our actual. If $E[T] > \theta$, we know that on average, we are overestimating the population parameter.

$T$ is an unbiased estimator of $\theta$ if the bias of $T$ is 0.

That is, if the $E[T] = \theta$

The example is that the $\bar{X}_n$ is an unbiased estimator of $\mu$.

**Example 6.1.3** (German Tank Numbers)

Imagine we have 3 tank numbers $413, 27, 518$
Assumptions:

- $n$ draws at random without replacement from $\{1, 2, 3, 4, \ldots, N\}$

Ideas:

- Take the sample max. Must be biased, bc always underestimates.
- $2\cdot$ sample average. This is biased, because overestimates

we are trying to estimate $N$.
Sample: $X_1, X_2, X_3, \ldots, X_n$ draws w/o replacements from $\{1, 2, 3, \ldots, N\}$

$$E[X_1] = \frac{N+1}{2} \text{(uniform dist fact)}$$

And then all the other expectations are the same!

$$E(\bar{x}_n = \frac{N+1}{2}$$

$$\boxed{2\bar{X}_n - 1 = \mu}$$

this is an unbiased estimator of $N$
Alternatively,
We take $M$ as the sample max. We imagine like seeing some tanks is like a deck of cards. We see 3 speicifc tanks, or 3 specific cards. We can take the runs between each seen tanks.
Or Number of visible gaps $= n$. The total length of the gaps $= M - n$. Where $n$ is the sample tanks. The average length of a gap $= \frac{M-n}{n}$.
Proposed estimator: $T_2 = M + \frac{M-n}{2} = M(1 + \frac{1}{n}) - 1 = M(\frac{n+1}{n} - 1)$
We start at the sample max, and see the average length of gap.
Is this unbiased?

$$E(\text{gap length}) = \frac{N-n}{n+1}$$

$$M = N - \text{gap}$$

# Chapter 7

## 7.1 Lecture 08: 12 February 2026

### 7.1.1 Probability, Expectation, Conditioning

**Example 7.1.1**

Players $A$ and $B$ take turns rolling a die. The first person that gets a 6 wins. Let Person A go first. Find

$P(A\text{wins})$. This answer should be bigger than a half, because the first person has an advantage.

Lets use a tree diagram:
If A gets a 6 right away, the game ends. Otherwise, B gets to go. This only continues if B also loses, and you go back to the start.
This is a recursive tree, therefore we can solve it quite easily. Thus we can condition on the first few moves.
Let $P(A\text{wins}) = x$
Then $x = \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot x$
This is a linear equation in $x$
In general terms:

$$\begin{aligned} P(A\text{wins}) &= p + q \cdot q \cdot P(A\text{wins}) \\ &= p + q^2 P(A\text{wins}) \\ &= \frac{6}{11} \end{aligned}$$

The intuition behind conditioning is natural. We have been doing it since high school.

**Example 7.1.2**

You have a class with 3 sections:
Section 1,2,3; with quiz averages of 80,70,90. What is the Class average? As stated, this question isn't answerable. You need the size (weight) of the sections.
If section 1 has 20, 2 has 40, 3 has 40. Then you can fully find the class average.

$$\text{avg} = 80 \cdot .2 + .4 \cdot 70 + 90 \cdot .4$$

This is an expectation. Value times probability. Each section's quiz probability is the conditional expectation of a student's score GIVEN that they are in section 1.

Conditional Expectation:
Pick a student at random, Let $T$ be their score and $S$ their section.

then:

$$E[T \mid S = 1] = 80$$
$$E[T \mid S = 2] = 70$$
$$E[T \mid S = 3] = 90$$

The abstraction here is that a student is a $\omega$ from $\Omega$. For any $\omega$, their group is either 1,2,3. That group has an average. We have defined a new random variable, $E[T \mid S]$.

The values of $E[T \mid S]$ are 70, 80, and 90. The probability of 70 is 0.4, 80 is 0.2, 90 is .4. This comes from the population size in each of the sections.

Therefore:

$$E[T] = E[E[T \mid S]]$$

This method of finding expectations are iterated expectations.

**Definition 7.1.1: Method of Iterated Expectations**

$$\begin{aligned}
E[T] &= \sum_{\forall T} t \cdot P(T = t) \\
&= \sum_{\forall t} t \sum_{\forall s} P(S = s, T = t) \\
&= \sum_{\forall t} \sum_{\forall s} t \cdot P(S = s) P(T = t \mid S = s) \\
&= \sum_{\forall s} \left[ \sum_{\forall t} t P(t = t \mid S = s) \right] P(S = s) \\
&= \sum_{\forall s} E[T \mid S = s] P(S = s) \\
&= E[E[T \mid S]]
\end{aligned}$$

### 7.1.2 Examples

**Example 7.1.3**

Random Sum. $X_1, X_2, \ldots$ i.i.d. with mean $\mu_x$ $N$ nonnegative integer valued, wiht a mean of $\mu_m$.
$S_N = X_1 + X_2 + \cdots + X_n$ if $N = n$, Let 0 if $N = 0$

Find $E[S_n]$, or the expected total.

If we knew the value of N, we know that the total is just $n\mu_x$

$$E[S_n \mid N = n] = n\mu_x \quad \text{(equality of numbers)}$$

$$E[S_n \mid N] = N\mu_x \quad \text{(equality of random variables)}$$

$$
\begin{aligned}
E[S_n] &= E[E[S_n \mid N]] \\
&= E[N\mu_x] \\
&= \mu_x E[N] \\
&= \mu_x \cdot \mu_N
\end{aligned}
$$

This is natural because we expect there to be $\mu_n$ terms, with each term being about $\mu_x$ big.

**Example 7.1.4** (Proving Geometric Expectation)

$$X \sim geometric(p) \text{ on } \{1, 2, 3, \dots\}$$

We can find the expectation of $X$ with conditional expectations.
R is the remainder.

$$X = 1 + R$$

$$E[X] = 1 + E[R]$$

$$E[R] = p(0) + qE[X]$$

$$
\begin{aligned}
E[X] &= 1 + E[R] \\
&= 1 + E[X]q \\
(1-q)E[X] &= 1 \\
E[X] &= \frac{1}{1-q} \\
&= \frac{1}{p}
\end{aligned}
$$

**Example 7.1.5**

Let $X = W_h$ be the waiting time for 1 head. $W_{HH}$ be the waiting times till $HH$.
To get to $W_{hh} = W_H + S$

$$E[W_{HH}] = E[W_H] + E[S] = \frac{1}{p} + E[S]$$

$$
\begin{aligned}
E[W_{HH}] &= \frac{1}{p} + E[S] \\
&= \frac{1}{p} + \big(1 \cdot p + q \cdot (1 + E[W_{HH}])\big) \\
&= \frac{\frac{1}{p} + 1}{1 - q} \\
&= \frac{\frac{1}{p} + 1}{p} \\
&= \frac{1}{p}^{2} + \frac{1}{p}
\end{aligned}
$$

# Chapter 8

## 8.1 Lecture 09: On Ipad

# Chapter 9

## 9.1 Lecture 10: 17 February 2026

Avoidable problems on Exams:

- Any probability exam: READ THE DANG QUESTION
- Just start writing outcomes. HHHHT. THTHT.
- do not do algebraic simplification unless problem explicitly tells you. If it explicitly says to find an answer, the calculation is straightforward, or easy to simplify.
  - finite sums are okay to keep as is.
  - infinite sums do need to simplify. Or some other finite sums.
- LOOK AT GRAPHS. Do you know how clustering of CR problem. Graph of birthday/matching problem.
- DO NOT LEAVE THINGS BLANK
- Can circle answers and draw arrows for multiple parts

Purpose of the upcoming 3 lectures is to show how crude models can give you a good answer. We are going to explore somewhat artificial settings. The point is to develop theory for finite numbers and calculations to see how you would break a mono alphabetic substitution code.

### 9.1.1 Describing the Model

Second example of a Stochastic process. The first example was the Chinese Restaurant problem.

Random walk example: foundation of all financial mathematics, physics, chemistry,etc.

Start at a point 0, move one step to the right with $p$ and to the left with $1-p$.

$$X_0, X_1, X_2, \ldots \text{stochastic process}$$

where $X_0$ is the random starting point.

The State Space $S$: the set of possible values of the $X_i$s. If you start at 0, and move either left or right, then all the integers are the states.

Because there is a sense of time,

$$X_0, X_1, \ldots, X_{n-1}, X_n, X_{n+1}, X_{n+2}, \ldots$$

time $X_n$ is called the present. Therefore, anything before is the past, and anything after is the future.

**Definition 9.1.1: Markov Property**

Given the state today, the distribution of where you are tomorrow, depends only on your current state, irrespective of how you got there.

More formally, Given the state today, the state tomorrow doesn't depend on the past. The future and the past are conditionally independent given the present.

$$P(X_{n+1} = i_{n+1} \mid X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}, X_n = i_n) = P(X_{n+1} = i_{n+1} \mid X_n = i_n)$$

This is known as a *one step transition probability.*

**Definition 9.1.2: Stationary Transition Probability**

$$P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i) \forall n$$

If you know you are at state $i$ at time $n$, is the same thing as if you were at state $i$ at time 0, because the past does not matter.

**Example 9.1.1** (Markov Chain)

5 State Chain, Look in textbook for the chain represented in this lecture. this is a transition diagram.
There is a .5 probability to stay at the same state, or 0.25 to go up and .25 to go down.
This is a *sticky*, because you can stay at the same node, *reflecting*, because you can't stay in one state, random walk

You can also describe this as a matrix:

$$p = \begin{bmatrix} .5 & .5 & 0 & 0 & 0 \\ .25 & .5 & .25 & 0 & 0 \\ 0 & .25 & .5 & .25 & 0 \\ 0 & 0 & .25 & .5 & .25 \\ 0 & 0 & 0 & .5 & .5 \end{bmatrix}$$

A row is a probability distribution of $P(X_{n+1} = x \mid X_n = i)$
The paths of this very simple model is varied. That means we can actually model stochastic processes.

*n-step transition probability*

$$P(X_n = j \mid X_0 = i) = P_n(i, j)$$

The chance you go from $i$ to $j$ in $n$ steps.
One step: $P(X_i = j \mid X_0 = i)$. This is what is in each row of $p$. For short this is $P_1(i, j)$

Let $n = 2$, find $P_2(i, j) = P(X_2 = j \mid X_0 = i)$
We can only partition according to $X_1 = k$

$$\begin{aligned} P_2(i, j) &= \sum_{k=1}^{5} P(X_1 = k, X_2 = j \mid X_0 = i) \\ &= \sum_{\forall k} P(X_1 = k \mid X_0 = i) \cdot P(X_2 = j \mid X_0 = i, X_1 = k) \\ &= \sum_{\forall k} P(X_1 = k \mid X_0 = i) \cdot P(X_2 = j \mid X_1 = k) \end{aligned}$$

This expression is a dot product. A row vector multiplied by a column vector.
Thus, $p^2(i, j)$, is the probability of getting from $i \to j$ in 2 moves.

In general: n-step transition matrix is $p^n$

**Definition 9.1.3: Steady State**

If you run a Markov chain for a long time, then it will forget the initial condition. This makes intuitive sense because of the Markov property. This only works if all nodes are communicative with each other.

Each conditional distribution converges to the same distribution as $n \to \infty$
The distribution of where you are at time $n$ does not matter of where you start, if n is large.

The steady state distribution is useful to tell if the distribution had a unique mode.

The chains we will study are irreducible. This means that we can get from everywhere to everywhere else.

# Chapter 10

# 11

## 10.1 Lecture 10: 19 February 2026

### 10.1.1 Midterm Solutions

**Example 10.1.1** (Problem 1)

a.) P(At least 3 vacant seats) $= v_3 + v_4 + v_5$

b.) Given that more than 2 people board, find probability that there are at least enough vacant seats for them.
people: 3 or 4.
3 people: at least 3 seats $((a))$ 4 people: at least 4 seats $(v_4 + v_5)$

$$\frac{p_3(a) + p_4(v_4 + v_5)}{p_3 + p_4}$$

**Example 10.1.2** (Problem 6)

$X \sim \text{Binom}(n, p)$. find $E[5^X]$

$$E[X^X] = \sum_0^\infty 5^k \cdot \binom{n}{k} p^k (1-p)^{n-k}$$

b.) elimate what cannot happen. it can only be binomial sum

$$\begin{aligned} E[5^X] &= \sum_{k=0}^{n} \binom{n}{k} (5p)^k (1-p)^{n-k} \\ &= (5p + (1-p))^n \\ &= (4p+1)^n \\ &= 2^8 = 256 \end{aligned}$$

**Example 10.1.3** (Problem 8b)

$D$ number of cards dealt till 4th ace.

$$E[D] = 4 + \sum_{n=4}^{51} \sum_{k=0}^{3} \frac{\binom{3}{k}\binom{48}{n-k}}{\binom{52}{n}}$$

We have a lot of sums. There could be 2 ways: tail sum or method of indicators.
Tail sum:

$$E[D] = \sum_{n=0}^{\infty} P(D > n)$$

Then the sum is somewhat like P(D¿n)

Tldr: Learn the properties of these concepts. Learn the general, then find the particular solution. Term by Term analysis is what is rewarded in this class. Build the answer. Take one problem and mess with it.

## 10.2 Markov Chains 2: Electric Booglaloo

$X_0, X_1, X_2, \ldots$ Are a Markov Chain, transition matrix $\mathbb{P}$

Finite state space (chains where there are a limit).

Irreducible: positive probability of getting from $j$ to $i$ for every $i, j$ via some path. One good way to establish this is to start a particular state, visit every other state, and comes back to A.

Periodic means no cyclical patterns, i.e. Only be on odd or even state. One way is to look along the diagonal of the chain, and if any of those are positive, then there will not be cyclical patterns.

**Theorem 10.2.1** Big Theorem

For every $i, j$,

$$P_n(i, j) \to \pi(j) \text{ as } n \to \infty$$

Where, $\pi(j) > 0$, $\sum_j \pi(j) = 1$
This is a probability distribution, called the steady state distribution of the chain.
This is because it doesn't matter where you start. The chance you end up at $j$ stays the same.

Consequences:

1. $\pi = [\pi(1)\pi(2)\ldots\pi(s)]$, Where $\pi(j)$ is the probability of being in state $j$.
2. Balance equations: $\pi$ solves the equations: $\pi = \pi P$. It is equations because $\pi$ is a vector, thus having $s$ equations and $s$ unknowns.
3. $\pi$ is the unique probability distribution that solves (2)

*Proof.*

$$P_{n+1}(i, j) = \sum_{\forall k} P_n(i, k) P_1(k, j)$$

$$\pi(j) = \sum_{\forall k} \pi(k) P(k, j) \quad n \to \infty$$

Since $\pi = [\pi(1)\pi(2)\ldots\pi(j)\ldots\pi(s)]$, we multiply $\pi \cdot P$. This gives a $1xs$ matrix, which is equivalent to what we just proved above.

**Definition 10.2.1: Why "Balance"?**

Imagine we are all running the chain. At every step, our chain moves from one state to another. Our moves are synced in that we transition at the same time.

LHS: Think of $\pi(j)$ as the proportion of particles leaving $j$.

RHS: $\sum_{\forall k} \pi(k)P(k,j)$. The interpretation is the proportion of particles leaving $k$, and then ending up in $j$

The steady state is can be interpreted as the proportion leaving $j$ is the same as the total proportion of particles entering $j$ .

**Detailed Balance**: For every pair $i \neq j$.
Suppose

$$\pi(i)P(i,j) = \pi(j)P(j,i) \implies \text{balance}$$

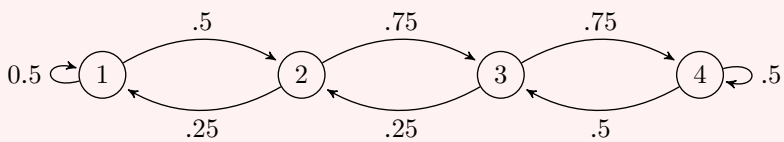



Figure 10.1:

$$\begin{aligned}
\pi(1).5 = \pi(2).25 &\implies \pi(2) = 2\pi(1) \\
\pi(2).75 = \pi(3) &\implies \pi(3) = 3\pi(2) = 6\pi(1) \\
\pi(3).75 = \pi(4).5 &\implies \pi(4) = 1.5\pi(3) = 9\pi(1)
\end{aligned}$$

$$\pi = [\pi(1)\ 2\pi(1)\ 6\pi(1)\ 9\pi(1)]$$

Sum is $18\pi(1) = 1 \implies \pi(1) = \frac{1}{18}$

# Chapter 11

## 11.1 Lecture 11: 24 February 2026

Detailed Balance: For all $i \neq j$, $\pi(i)P(i,j) = \pi(j)P(j,i)$.

If a probability distribution $\pi$ satisfies these, then it is the steady state distribution of the chain. In other words, if you run the distribution a long time, then it will just be $\pi$.

Detailed balance: $\frac{\pi(i)}{\pi(j)} = \frac{P(i,j)}{P(j,i)}$

All of our walks have been on a small process. We can build this connection through examples, especially with MKC on text. Markov's first chain was looking at text and coding 1 and 0 for vowel vs constant. Look at nature of trials between trials and constants.

**Example 11.1.1** (Small Markov Chain)

Let us start with a small alphabet, of 3 letters. ADT as our letters.
We have a message of length 4 $atdt$. We have to decode this thing, where there are 6 different possible decoders because our alphabet has 3 letters.

We can use a really simple and crude model of assuming English is a one step markov chain on the letters. We have transition probabilities of $a \to d, a \to t, \ldots$

Model is not trying to have a complete truth, just model enough randomness. With our trivial example, we enumerated all the decoders, but that wouldn't be possible with a full alphabet. We want a process that finds decoders with high scores.

We could do a markov chain on the decoder. If we are at one decoder, the transition to the next decoder only matters on where I am currently.

### 11.1.1 General alphabet

Each decoder has a score. "normalize the scores to create a distribution $\pi$". We can't actually do this, but this is the general idea in the abstract.

**Example 11.1.2**

Target: We want the decoder with the highest score, that is, the mode of $\pi$.

Idea: Construct a MKC on the decoders, s.t. the MKC has steady states $\pi$
If we run the chain for a long time. Then we are likely to see the mode here. Observed value after a long time should be the mode of $\pi$

States: All possible decoders. This is finite and countable. Start with any symmetric ($Q(i,j) = Q(j,i)$) irreducible transition matrix $Q$. (Proposal Matrix)
Create transition behavior for $X_0, X_1, X_2, \ldots$, where each state is a decoder.
Given $x_n = i$
1.) Go to $i$ th row of $Q$. Here we see a conditional distribution. Pick a state $j$ according to the distribution in this row. $Q$ proposes a move to $j$
2.) Ratio $r(i,j) = \frac{\pi(j)}{\pi(i)}$. We look if this ratio is $r \geq 1$, then we move to that state $j$. If $r(i,j) < 1$, toss a coin that lands heads with chance $r(i,j)$. If heads, move to $j$, otherwise stay at $i$.

Case 1: $\pi(i) = \pi(j)$

$$\begin{aligned}
P(i,j) &= Q(i,j) \cdot 1 \\
P(j,i) &= Q(j,i) \cdot 1 \\
&= Q(i,j) \\
&= P(i,j) \\
\pi(i)P(i,j) &= \pi(j)P(j,i)
\end{aligned}$$

Therefore, this case works.
Case 2: $\pi(j) < \pi(i)$

$$\begin{aligned}
P(i,j) &= Q(i,j) \cdot r(i,j) \\
&= Q(i,j)\frac{\pi(j)}{\pi(i)} = Q(j,i)\frac{\pi(j)}{\pi(i)} \\
P(j,i) &= Q(j,i) \cdot 1 \\
\pi(i)P(i,j) &= \pi(j)P(j,i)
\end{aligned}$$

Which also satisfies the detailed balanced equations.

# Chapter 12

## 12.1 Lecture 11: 26 February 2026

If we have $X$ expectation $\mu_X$. Deviation $D = X - \mu_X$

What is the expectation of the deviation? $E[D] = E[X - \mu_X] = E[X] - \mu_X = 0$ This means that on average, the deviations cancel each other out.

Therefore, we are not interested in the average deviation, we are instead interested in the average distance of the deviation. We can use absolute values here, however they are hard to work with because its not differentiable at 0. Instead, we will use $x^2$ instead.

$$E[D^2] = E[(X - \mu_X)^2] = \mathrm{Var}(X)$$

The definition of variance is the mean squared deviation from $X$. Variance is usually positive, unless $X = 0$. The unfortunate part about variance is that we have messed up our units; it will always be in squared units. Therefore we take the square root to get standard deviation.

$$SD(X) = \sqrt{\mathrm{Var}(X)}$$

How do we calculate variance?

$$\begin{aligned}\mathrm{Var}(X) &= E[(X - \mu_X)^2] \\ &= \sum_{\forall x} (x - \mu_x)^2 \cdot P(X = x)\end{aligned}$$

If you give prof A. A negative stave, don't ask for partial credit.....

Properties:

1. Linear Function: $Y = aX + b, \mu_y = a\mu_x + b,\ \sigma_y = |a|\sigma_x$

$$D_y = (aX + b_) - (a\mu_x + b) = aD_x$$

$$\mathrm{Var}(Y) = a^2\,\mathrm{Var}(X)$$

2. The variance is the expectation of the square - the square of the expectation.

$$\begin{aligned}\mathrm{Var}(X) &= E[(X-\mu_x)^2] \\ &= E[X^{2-2\mu_x X+\mu_x^2}] \\ &= E[X^2] - 2\mu_x E[X] + \mu_X^2 \\ &= E[X^2] - 2\mu_X \\ &= E[X^2] - \mu_X^2\end{aligned}$$

**Definition 12.1.1: Standard Units**

$$Z = \frac{X-\mu_x}{\sigma_x}$$

$$E[Z] = 0$$

$$SD(Z) = 1$$

**Example 12.1.1** (Indicators)

If we have indicator $I = \begin{cases} 1, p \\ 0, \text{otherwise} \end{cases}$

$\mu_I = p$

Therefore $E[I^2] = E[I]$, since the values of $I^2$ and $I$ are the same.

$$\mathrm{Var}(I) = p - p^2 = p(1-p)$$

**Example 12.1.2** (Uniform)

$X$ uniform on $1, 2, 3, \ldots, n$

$$E[X] = \frac{n+1}{2}$$

$$E[X^2] = \sum_{k=1}^{n} k^2 \frac{1}{n} = \frac{1}{n}\sum_{k=1}^{n} k^2$$

$$= \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6}$$

$$\mathrm{Var}(X) = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{n^2-1}{12}$$

**Example 12.1.3** (Poisson)

$X \sim \mathrm{Poisson}(\mu)$

$$E[X] = \mu$$

$$E[X(X-1)] = \mu^2 = E[X^2 - X] = E[X^2] - E^X$$

$$E[X^2] = \mu^2 + \mu$$

$$\text{Var}(X) = \mu^2 + \mu - \mu^2 = \mu$$

### 12.1.1 Interpretations

Bounds:

1. Let $X \geq 0$. Suppose we know $\mu_x$. $P(X \geq c)$?

   Suppose $g$ and $h$ are two functions on the real line, with $g \geq h$. So $E[g(X)] \geq E[h(X)]$.

   $$P(X \geq c) = E[I(X \geq C)]$$

   We can bound $I$ by $g(x) = \frac{1}{c}X$

   Since $g \geq h$, then $E[g(x)] \geq E[h(x)]$, which means that $E[\frac{1}{c}X] \geq E[I(X \geq c)] \implies E[X]/c \geq P(X \geq c)$

   Thus this is Markov's inequality.

   $$P(X \geq c) \leq \frac{E[X]}{c}$$

2. $P(|X - \mu_x| \geq c) = P((X - \mu_x)^2 \geq c^2) \leq E[(X - \mu_x)^2]/c^2 = \frac{\text{Var}(X)}{c^2}$

   This is the Kuibyshev's Inequality.

   In another way:

   $$P(|X - \mu_x| \geq k\sigma_x) \leq \frac{1}{k^2}$$

# Chapter 13

## 13.1 Lecture 12: 3 March 2026

$$\begin{aligned} X &: \mu_X, \sigma_x^2 \\ Y &: \mu_Y, \sigma_Y^2 \\ S = X + Y \mu_S &= \mu_X + \mu_Y \end{aligned}$$

but what is $\sigma_S^2 = \mathrm{Var}(S)$

$$\begin{aligned} \mathrm{Var}(S) &= E[D_s^2] \\ D_S &= S - \mu_s = X + Y - (\mu_x + \mu_y) = D_x + D_y \\ E[D_S^2] &= E[(D_X + D_Y)^2] \\ &= E[D_X^2] + 2E[D_X D_Y] + E[Y^2] \\ &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y) \end{aligned}$$

In general, the variance of the sum is not the sum of the variances, because of the co variance term.

$$E[D_x D_y] = E[(X - \mu_x)(Y - \mu_Y] = \mathrm{Cov}(X, Y)$$

Co variance has two principle uses.

The units is units of $X\cdot$ units of $Y$.

1. Get rid of the units entirely: $\frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$. This is the $E\left[\frac{X-\mu_x}{\sigma_x} \cdot \frac{Y-\mu_Y}{\sigma_Y}\right]$, which are the average of the two product of the standard units, or in other terms, the correlation between X and Y.
2. Finding the variance of sums.

### 13.1.1 Properties of Cov

1. $\mathrm{Cov}(X, c) = 0$
2. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
3. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$ (Special case of 4)
4. $\mathrm{Cov}(X, Y) = E\left[XY - X\mu_y - Y\mu_X + \mu_x\mu_y\right] = E[XY] - 2\mu_x\mu_y + \mu_x\mu_y = E[XY] - \mu_x\mu_y$.
   In general, $E[XY] \neq \mu_x\mu_y$
5. $X \perp Y \implies \mathrm{Cov}(X, Y) = 0$. This is really important when finding the sum of variances, especially $X \perp Y$

6. $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$
7. $\text{Cov}(cX, wY) = cw\,\text{Cov}(X, Y)$

**Example 13.1.1** (Application of property 4)
$I_A$ indicator of A, $I_B$ indicator of B.

$$E[I_A] = P(A), \quad \text{Var}(I_A) = P(A)(1 - P(A))$$

$$\begin{aligned}
\text{Cov}(I_A, I_B) &= E[I_A I_B] - P(A)P(B) \\
&= E[I(A \cap B)] - P(A)P(B) \\
&= P(AB) - P(A)P(B) \\
&= P(A)P(B \mid A) - P(A)P(B) \\
&= P(A)(P(B \mid A) - P(B))
\end{aligned}$$

If the sign of the covariance is positive, then B given A is greater than the probability of B. This is called positive association, as B is more likely to happen if A happens. In other words, if $\text{Cov}(I_A, I_B) > 0$, $P(B \mid A) > P(B)$.
Similarity, if $\text{Cov}(I_A, I_B) < 0, P(B \mid A) < P(B)$ , which is negative association.

**Example 13.1.2** (Is cov ever 0?)

$$E[XY] = \sum_{\forall x} \sum_{\forall Y} xyP(X = x, Y = y)$$

If x and y are independent, then chance of the intersection becomes a product of the two chances.

$$\begin{aligned}
&= \sum_{\forall x} \sum_{\forall y} xyP(X = x)P(Y = y) \\
&= \sum_{\forall x} xP(X = x) \sum_{\forall Y}^{Y=y} \\
&= E[X]E[Y] \\
&= \mu_x \mu_y
\end{aligned}$$

Thus we see if and only if X and Y are independent, then the covariance of X and Y is 0.

**Example 13.1.3** (Applying Covariance Properties)
**Bilinearity of Covariance.**

$$\begin{aligned}
\text{Cov}(X - 4Y + 1, 2X + 3Y - 15) &= \text{Cov}(X - 4Y, 2X + 3Y) \\
&= 2\,\text{Cov}(X, X) + \text{Cov}(X, 3Y) - 8\,\text{Cov}(X, Y) - 12\,\text{Cov}(Y, Y) \\
&= 2\,\text{Var}(X) + -5\,\text{Cov}(X, Y) + 12\,\text{Var}(Y)
\end{aligned}$$

### 13.1.2 General Formula for variance of sums

$$\begin{aligned}\mathrm{Var}(\sum_{i=1}^{n} X_i) &= \mathrm{Cov}(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(X_i, X_j) \\ &= \sum_{i=1}^{n} \mathrm{Var}(X_i) + \sum_{i \le i \ne j \le n} \sum \mathrm{Cov}(X_i, X_j)\end{aligned}$$

$X_1, X_2, \ldots, X_n$ are i.i.d. with mean $\mu$, SD $\sigma$

$$S_n = \sum_{i=1}^{n} X_i, \quad E[S_n] = n\mu$$
$$\mathrm{Var}(S_n) = n\sigma^2, \quad SD(S_{n)} = \sqrt{n}\sigma$$

Take $\bar{X}$, $E[\bar{X}] = \mu$ and $\mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ We see that the sample average becomes a spike, which makes it a very good estimator.

**Example 13.1.4** (SD of binomial)
$X \sim \mathrm{Binom}(n, p)$, where $X = I_1 + I_2 + I_3 + \ldots$, where $I_j$ indicates success on Trial $j$.
$E[X] = np$

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{Var}(I_1) + \mathrm{Var}(I_2) + \ldots \quad \text{by independence} \\ &= np(1-p)\end{aligned}$$

SD $= \sqrt{npq}$

**Example 13.1.5** (SD of Poisson)
SD of Poisson is $\sqrt{\mu}$
If we have binomial, large $n$, small $p$, s.t $np \approx \mu$.
SD of the binomial is $\sqrt{npq}$. q is approx 1, and np is approx $\mu$. Thats how we get the sd of the poisson.

**Example 13.1.6** (SD of the hypergeometric)
$X_1, X_2, X_3, \ldots, X_n$ is a SRS from $N$ numbers that have mean $\mu$ and SD $\sigma$.
$S_n = X_1 + X_2 + \cdots + X_n$
By symmetry, $\mathrm{Var}(\sum_{\forall i} X_i) = n\sigma^2 + n(n-1)\,\mathrm{Cov}(X_1, X_2)$

For $1 \le n \le N$,

$$\mathrm{Var}(S_n) = n\sigma^2 + n(n-1)\,\mathrm{Cov}(X_1, X_2)$$

If $n = N$, which is a Census, the variance is 0.

$$0 = \mathrm{Var}(S_n) = N\sigma^2 + (N)(N-1)\,\mathrm{Cov}(X_1, X_2)$$

Thus

$$\mathrm{Cov}(X_1, X_2) = \frac{\sigma^2}{N-1}$$

In general, for a sample size $n$,

$$\begin{aligned}\mathrm{Var}(S_n) &= n\sigma^2 + n(n-1)\left(\frac{-\sigma^2}{N-1}\right) \\ &= n\sigma^2\left(1 - \frac{n-1}{N-1}\right)\end{aligned}$$

This says that if we sample without replacement, the error is lower. This fraction is pretty close to 0.
Varaince of the hypergeometric:

$$n \cdot \frac{G}{N} \cdot \frac{B}{N} \cdot \left(1 - \frac{n-1}{N-1}\right)$$